Neha Sharma

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Education

The National Institute of Engineering CGPA: 9.1/10
B.Tech. in Computer Science and Engineering *Aug 2023 – Aug 2027*

Relevant Coursework:
Data Structures and Algorithms, Operating Systems, DBMS, OOPS, Computer Networks

Rainbow School Percentage: 95.2%
Intermediate *2022 – 2023*

Rainbow School Percentage: 97.2%
High School *2020 – 2021*

Skills and Competencies

Languages: C, C++, Python, Java, SQL, HTML/CSS
Developer Tools: Git, GitHub
Libraries: Pandas, Matplotlib, Numpy, Scikit-learn, TensorFlow, Pytorch

Experience

Indian Institute of Technology, Roorkee Roorkee | On Site
Summer Research Intern - AMSC | Deep Learning, CV July 2025 – August 2025

- Research Advisor : Dr. Millie Pant (HoD- AMSC)
- Developed a Deep Learning model for Polyp Segmentation in Endoscopic Colonoscopy Frames using DeepLabV3+ and Unet
- Achieved an IoU score of 0.97 and Dice Loss of 0.042
- Enhanced model interpretability by implementing Explainable AI (XAI) techniques for transparent decision-making.
- Utilized the Kaggle Endoscopic Colonoscopy dataset for model training, testing and performance evaluation.

Projects

Word Sense Disambiguation in Indic Languages (NLP Research Project) April 2025

- Developed a Word Sense Disambiguation (WSD) system for Hindi by combining text preprocessing, semantic feature extraction, Word2Vec embeddings, and machine learning classifiers.
- Developed a custom Hindi stopword library for efficient semantic filtering in low-resource NLP tasks.

Targeted Intervention Optimization using Uplift Modeling and Causal Inference January 2026

- Designed an end-to-end experimentation framework and built uplift models to estimate individual treatment effects, enabling targeted interventions over blanket policies.
- Implemented causal meta-learners (T-Learner / X-Learner) and evaluated using Qini and AUUC metrics, improving incremental response capture over standard propensity models.
- Identified and prioritized "persuadable" segments to maximize net lift while reducing wasted treatments on sure-things and lost-causes.

Retrieval-Augmented Generation System with Orchestrated LLM Workflows December 2025

- Designed and deployed a Retrieval-Augmented Generation (RAG) pipeline using LangChain and vector search to ground LLM responses in domain-specific documents.
- Orchestrated multi-step reasoning workflows with LangGraph, enabling tool-calling, memory, and controlled agent execution for complex query resolution.
- Optimized retrieval and generation quality through embedding selection, chunking strategies, and reranking, improving factual accuracy and reducing hallucinations.

Achievements & Certifications

1st Prize — Data Sprint 2025 (FinTech ML Hackathon)

- Built an end-to-end Credit Risk Prediction system using raw FinTech data.
- Engineered repayment-based default labels; addressed class imbalance using SMOTE and class weighting.
- Trained Random Forest, Balanced RF, and XGBoost models with threshold tuning to maximize recall.

Machine Learning Specialization — Andrew Ng (Coursera)
Deep Learning Specialization — Andrew Ng (Coursera)